

WORLD TRIATHLON

WATER QUALITY STATEMENT

Triathletes are very intense exposed to the contaminated water during training and triathlon competitions. They can become infected due to water ingestion and or due to skin or mucous membrane contact with the water. (Schep G. 1992) It is very plausible that, during a competition, some athletes are more prone to develop infections than under normal circumstances. There are several physiological mechanisms during exercise that may lower the resistance to infections during and shortly after an endurance effort like a triathlon. It is mainly based upon three mechanisms:

- the hormonal stress reaction
- the drop in some specific immune parameters like the concentration and activity of natural killer cells
- the possible failure of local protective mechanisms of the gastrointestinal tract and of the skin.

1. Water quality problems

The most important health problem due to contaminated swimming water is associated with microbiological contamination.

For triathlon the most relevant pathogens are in order of importance associated with the following risk factors:

1.1 Human and animal faecal contamination

There is an association with the following pathogens: enterococci/faecal streptococci, E.Coli, faecal coliformi, total coliform, salmonella, shigella, campylobacter, vibrio cholera, gastro enteritis viruses, enteroviruses, parasitic protozoa, some other exotic parasites, Aeromonas and Plesiomonas.

1.2 The occurrence of rodents

Most relevant are brown rats and muskdeer, who can pass leptospirae in their urine into the water when they are infected. A determination of leptospirae in surface water is very complicated and not feasible. The most important reason is the very low dose that is necessary for an infection (below the limit of laboratory detection) and the fact that the concentration of leptospirae in the water is almost always extremely low.

1.3 Eutrofication of the water

Some pathogenic bacteria like *Aeromonas*, *Plesiomonas*, *Cyanobacteria*, can actually multiply themselves in surface water. This multiplication is stimulated if the water contains more nutritious matter (eutrofication). People may be exposed to Blue-Green algae or Red Tide algae toxins by drinking or bathing in contaminated water.

1.4 Bird colonies

Since birds can be infected with *Plesiomonas*, *Campylobacter* and *Salmonellae* they may contaminated the water with those organisms.

1.5 Industrial cooling water installations

It is proven that naturally occurring and potential severely pathogenic amoeba's can grow in a hot environment like the plume of an industrial cooling installation.

2. Risk factors

The most important risk factor is faecal contamination due to human faeces. Animal faeces is less important in harbouring human pathogens. Human faeces can come into the water directly (bad sanitation, in many countries discharge of non purified sewage and by discharge from house boats). However also purification of sewage is not able to control all the disease causing organisms. Dependant on the flow and the extent of the discharge there can be large fluctuations in the quality of the water.

Faecal contamination can harbour many potentially pathogens. It is not feasible to test all those micro organisms. Further on there is also a lack of data considering the quantitative relationship of the concentration of many of those pathogens and the risks of disease. Especially the viruses are the most hazardous.

3. The commonly used indicator.

Bacterial indicators of faecal contamination considered are enterococci/faecal streptococci, *E.Coli*, faecal coliformi and total coliform. Thermotolerant coliform bacteria are present in human and animal faeces. However they also occur in other environments that bear no relationship to faecal contamination (discharge from paper industry, brewery's etc). *Escherichia coli* (*E.Coli*) is a species of faecal coliform that is specific for faecal material from humans and other warm-blooded animals. Enterococci are a subgroup within the faecal streptococcus group and are distinguished by their ability to survive in salt water. Faecal coliforms as a group were determined to be a poor indicator of the risk of GI illness (US EPA 2002).

4. Water quality control

Water quality control consists of:

4.1 Visual inspection

Quite obviously measurable entities that are not however so specific indicators of the water quality, like:

- the colour of the water does not change in an abnormal way
- the absence of oils that float and smell of phenol
- a transparency of the water of more than one meter.(this may only be less when natural circumstances cause this)
- the absence of the Blue-Green Algal Bloom or the Red Tide Algal- Bloom
- the absence of rodents
- the absence of bird colonies
- the absence of discharge of cooling water from factory
- the absence of garbage
- the absence of sewage discharge or the occurrence of house and boats

4.2 Bacterial and or viral tests

Those tests are a measure for faecal contamination and they are very important regarding swimming associated health risks. Traditionally the microbiological quality of waters has been measured by the analysis of indicator microorganisms. Human enteric virus are the most likely pathogens responsible for waterborne diseases from recreational water, but detection methods are complex and costly for routine monitoring and so the main parameters analysed for compliance with the Directive are indicator organisms.

EEA Standards

The **Directive 2006/7/EC** reduced the number of parameters from 19 to 2 key microbiological parameters and the faecal contamination is assessed by determination of two mandatory indicator bacteria:

- Escherichia Coli
- Enterococci

The choice of microbiological parameters is based on available scientific evidence provided by epidemiological studies conducted by the WHO and health Institutes in Germany, France and Netherlands.

For some characteristics E. Coli may be considered a more useful indicator than faecal coliforms and it has been included in all recent laws regarding fresh, marine and drinking waters.(Baudisova D.1997, Briancesco R. 2005) E.Coli and enterococci are adequate indicators of GI illness in marine water. In fresh water, E. Coli is a more reliable as consistent predictor of GI illness than is enterococci. E.Coli was superior to

enterococci at predicting illness and the E.Coli guideline level was supported because exposure below presented no significant risk, whereas exposure above were associated with an elevated and statistically significant increased risk of GI illness(Foulon et al 1983, Kay et al.1994, McBride et al.1998). The body of literature does support the use of enterococci and E.Coli as useful predictors of GI illness in marine environment. (Wade TJ et al. 2003) Also salmonellae and enteroviruses there are standards, however one will not determine those micro-organisms on a routine base.

- d) It is very difficult to establish the relationship with the exposure to the swim water because there is no routine testing for those viruses and because the symptoms in general only start several weeks after the exposure.
- e) No treatment is available for those viral illnesses so a practical working doctor is not so eager to find out which virus was the cause of the disease. In the contrary when a doctor considers a possible leptospirosis infection that may look quite similar it has much more practical implications for the treatment to determine whether or not the patient is infected. Heavy rainfall and flooding contribute to emergence of leptospirosis, implying that climate change might lead to a further increase (Brockmann et al., 2010).

We learn from this that it may very well be possible that the health problems associated with faecal contaminated swimming water do not restrict themselves to the rather innocent gastroenteritis.

6. World Triathlon Water Quality Rules

Since 2010 World Triathlon adopted the EEC standards 2006/7/EC of bathing water quality.

Especially for international competitions it is absolutely necessary to get a clear view on the water quality of such competitions. Since in some countries there is risk for some very serious disease causing health problems (Migliorini S,2009).

6.1 Water quality tests submit to World Triathlon

- According to World Triathlon rules, the LOC must submit water quality tests:
- Two months before the competition
- 7 days before the competition
- On the first competition day of the event for statistical purposes only.

Samples of the water collected from three different locations on the swim course will be separately analysed and the poorest results will determine if the swim can take place.

The following values are set as the World Triathlon in the different types of water.

Sea and Transitional waters:

- PH between 6 and 9
- Enterococci not more than 200 per 100 ml (ufc/100ml)
- E. Coli not more than 500 per 100 ml (ufc/100ml)
- Absence of positive visual evidence of Red Tide Algal bloom.

Inland waters

- PH between 6 and 9
- Enterococci not more than 400 per 100 ml(ufc/100ml)
- E. Coli not more than 1000 per 100 ml(ufc/100ml)
- The presence of Blue-Green Algal bloom/scum (cyanobacteria) with more than 100.000 cells/ml.

This test is only acquired in case of positive visual evidence of Blue Green Algal bloom. Because of the potential for rapid scum formation daily visual inspection is mandatory by the LOC Medical Director in the two weeks before the competition in the area prone to scum formation. Where not scums are visible, but the water shows strong greenish discoloration, turbidity and the transparency is less than 0.5 m the cyanobacteria test must be performed.

All units have to be cfu/100 ml or nmp/100 ml (cfu : Colony Forming Units; nmp : Most probable Number)

The water quality results delivery may vary from 48 hours to 96 hours depending on the methodology that the laboratory is using; therefore, making a decision on competition delay based purely on data that is not reflecting the current conditions 100% is not advisable. Particularly in cases where the event is running under raining conditions World Triathlon use approach in the cases but not limited to, recommended by WHO to guide decision making (Water Quality Decision Matrix) by combining:

- water quality analysis
- visual inspection
- weather forecasts

World Triathlon recommend to organise only Triathlon in swimming water that falls in the Decision Matrix Excellent (1) or Good Water Quality category (2).

The WT Technical Delegate and/or the WT Medical Delegate assign a classification of the water quality for the day according to the defined matrix, and communicate the results to the Event Water Quality Evaluation Panel.

The Event Water Quality Evaluation Panel is represented by :

- World Triathlon Medical Committee representative
- World Triathlon Medical Delegate (if not applicable the World Triathlon Technical Delegate)
- LOC Medical Director (if not applicable the LOC Race Director).

Decision making

If the water quality levels show values three (3) or four (4), the swim will proceed only if the Events' Water Quality Evaluation Panel permits with a simple majority. In this case, the local water quality regulations may be considered.

In case of discrepancy between the results we must consider that the E.Coli are the best indicator bacteria for Inland water and the Enterococci are the best indicator for Sea water.

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On behalf of the World Triathlon Medical Committee